

- 9 On 11 March 2011, Japan suffered a Magnitude 9.0 Earthquake off the eastern coast of Japan. One of the consequences of the Earthquake was a tsunami that crippled the Fukushima Daiichi nuclear power plant, which subsequently released significant amounts of radioactive materials and by-products into the atmosphere and environment.

Tables 9.1 and 9.2 show the surface activities of several radionuclides measured on 31 March 2011 from soil samples at several locations in and around the Fukushima district, their half-lives and decay modes.

Surface Activity (Bq m^{-2})						
Location	Tellurium-129m	Tellurium-132 / Iodine-132	Iodine-131	Caesium-134	Caesium-137	Barium-140 / Lanthanum-140
A	208 370	111 960	824 150	205 260	211 480	8 397
B	388 600	195 640	1 163 120	477 040	479 720	22 244
C	754 920	382 120	1 887 300	894 720	922 680	35 416
D	71 928	35 478	481 140	77 031	78 489	2 527

Table 9.1 Surface Activities of Radionuclides

Radio-nuclide	Tellurium-129m	Tellurium-132	Iodine-132	Iodine-131	Caesium-134	Caesium-137	Barium-140	Lanthanum-140
Half-life (Days)	33.6	3.20	2.30	8.02	755	11 000	12.8	1.68
Decay Constant (s^{-1})	2.39×10^{-7}	2.51×10^{-6}	3.49×10^{-6}		1.06×10^{-8}	7.29×10^{-10}	6.27×10^{-7}	4.78×10^{-6}
Decay Mode	γ	β^-	β^-	β^-	β^-	β^-	β^-	β^-

Table 9.2 Half-lives and Decay Modes of Radionuclides

34 Se selenium 79.0	35 Br bromine 79.9	36 Kr krypton 83.8	Notations A: Mass/Nucleon Number Z: Atomic/Proton Number X: Chemical Symbol Chemical Name Relative Atomic Mass Example of Nuclide Representation: A_ZX
52 Te tellurium 127.6	53 I iodine 126.9	54 Xe xenon 131.3	
84 Po polonium —	85 At astatine —	86 Rn radon —	

Fig. 9.1 Elements Around Iodine on the Periodic Table

(a) Using the data from **Table 9.1**, **Table 9.2** and **Fig. 9.1**,

- (i) calculate A_β , the total activity per m^2 due to beta decay, of the most contaminated location.

$$A_\beta = 382120 + 1887300 + 894720 + 922680 + 35416 \quad [1]$$

$$= 4.1222 \times 10^6 \text{ Bq m}^{-2} \quad [1]$$

$$A_\beta = \dots\dots\dots \text{Bq m}^{-2} \quad [1]$$

- (ii) for the location C,

1. calculate the decay constant of Iodine-131,

$$\lambda_i = \frac{t_{1/2}}{\ln 2}$$

$$\lambda_i = \frac{8.02 \times 24 \times 3600}{\ln 2} \quad [1]$$

$$\lambda_i = 1.00 \times 10^{-6}$$

$$\text{decay constant} = \dots\dots\dots \text{s}^{-1} \quad [1]$$

2. calculate N_i , the number of Iodine-131 nuclei present per m^2 ,

$$A_i = \lambda_i N_i$$

$$N_i = \frac{1887300}{1.00 \times 10^{-6}} \quad [1]$$

$$N_i = 1.89 \times 10^{12} \quad [1]$$

$$N_i = \dots\dots\dots [2]$$

3. calculate M_i , the mass of Iodine-131 nuclei present per m^2 ,

$$M_i = 1.887 \times 10^{12} \times 131 \times 1.66 \times 10^{-27}$$

$$M_i = 4.10 \times 10^{-13} \text{ kg} \quad [1]$$

$$M_i = \dots\dots\dots \text{kg} \quad [1]$$

4. calculate $A_{T,i}$, the total activity present due to Iodine-131 in 7.5 km^2 of the location.

$$A_{T,i} = 1887300 \times 7500000$$

$$A_{T,i} = 1.42 \times 10^{13} \text{ Bq}$$

$$A_i = \dots\dots\dots \text{Bq} \quad [1]$$

(v) write down the decay equation for:

1. Tellurium-129m



2. Tellurium-132



(b) Explain why after 16.1 days, another sample taken at the same location had significantly more than a quarter of the original activity due to Iodine-131.

25% of the original activity assumes that 2 half-lives has occurred for iodine-131 that is not replenished / with a fixed mass. [1] However, as iodine-131 decays to another nuclide, there could be other radioactive nuclide which also decays to iodine-131 but with a shorter half-life, hence significantly increasing the mass of iodine-131 present. [1].....[2]

(c) Among the radioactive fallout, Iodine-131 and Caesium-137 are major concerns as both elements can be absorbed by the human body through inhalation as well as through the consumption of contaminated food and water.

(i) Even though the activity of Iodine-131 is significantly greater than that of Caesium-137 in all locations tested, Caesium-137 will present a greater hazard in the long term. Explain why this is so.

The half-life of iodine-131 at only 8.02 days is short while the half-life of cesium-137 at 11000 days is significantly longer. In the long term, the activity due to iodine-131 would have reduced to a relatively insignificant level while the activity of cesium-137 would remain significant.[2]

(ii) Iodine is absorbed by the body and accumulates in the thyroid. Explain why Iodine-131 is dangerous when ingested.

Iodine-131 emits beta particles which is an ionising radiation which can damage and kill living cells. As it is accumulated in the thypoid after ingestion, it can cause severe and irreparable damage to the surrounding cells over a long period of time and significantly increase the chance of contracting cancer.[2]

- (iii) Suggest why the consumption of potassium iodide pills before Iodine-131 exposure may help reduce the chances of irreparable damage to the thyroid or contracting thyroid cancer.

The consumption of potassium iodide pills before exposure could saturate the thyroid with
 non-radioactive iodine. This reduces the accumulation of radioactive iodine in the thyroid,

 hence reducing the time it spends in the thyroid.[2]

- (d) Radioactive materials are often considered dangerous due to its high energy output with a relatively small mass and its hazardous nature to living things.

- (i) By considering the decay constants of the radioactive materials listed that undergoes beta decay, suggest and explain which material could be the most dangerous.

As Lanthanum-140 has the largest decay constant of $4.78 \times 10^{-6} \text{ s}^{-1}$, this suggests that the
 material has the highest percentage of nuclide undergoing radioactive decay. [1] This means

 that for the same mass and exposure duration, Lanthanum-140 produces the most ionising

 particles (beta) and could be releasing the highest amount of energy per unit time. [1] [2]

- (ii) Suggest and explain another physical factor that contributes to the dangers of radioactive material.

Type of radiation: alpha and beta particles are more ionising than gamma rays.

Half-life: longer half-life means longer periods of radioactivity

Activity: higher activity means higher energy output[2]

Binding energy per nucleon: higher BE/N = higher energy output per decay [Total: 20]